

Guideline Title: Neonatal Stabilization and Resuscitation

Guideline Number: GUID-3203-PED019

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- I. **PURPOSE:** To ensure neonatal stabilization and resuscitation while undergoing transition from intrauterine to extrauterine life. These recommendations are also applicable to neonates who have completed perinatal transition and require resuscitation during the first few weeks following birth.
- II. **Considerations for transport**
- III. **DEFINITIONS:**
- IV. **DEPARTMENT GUIDELINE:**
 - a. Stabilization
 - i. Determine the Estimated Gestational Age (EGA) based on last menstrual period or recent ultrasound and Estimated Date of Confinement (EDC). Ask Expected Due Date (EDD) or it can be estimated from the LMP
 - ii. Provide warmth to the infant.
 - iii. Assess and manage airway, breathing and circulation.
 - iv. If baby is a term gestation with good tone and normal breathing/crying, allow mother to hold while performing routine care
 - v. Perform 1 and 5 minute APGAR assessment
 1. An APGAR score is given for each sign at 1 minute and 5 minutes after the birth. If there are problems with the baby an additional score is given at 10 minutes.
 2. A score of 7 - 10 is considered normal, while 4 - 7 might require some resuscitative measures. APGARs of 3 and below requires immediate resuscitation. If neonatal resuscitation is required, follow Neonatal Resuscitation Program (NRP) guidelines.
 - a. Activity (Muscle Tone):
 - i. 0 - Limp; no movement.
 - ii. 1 - Some flexion of arms and legs.
 - iii. 2 - Active motion.
 - b. Pulse (Heart Rate):
 - i. 0 - No heart rate.
 - ii. 1 - Fewer than 100 beats per minute.
 - iii. 2 - At least 100 beats per minute.
 - c. Grimace (Reflex Response):

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- i. 0 - No response to airways being suctioned.
 - ii. 1 - Grimace during suctioning.
 - iii. 2 - Grimace and pull away, cough, or sneeze during suctioning.
 - d. Appearance (Color):
 - i. 0 - The baby's whole body is completely bluish-gray or pale.
 - ii. 1 - Good color in body with bluish hands or feet.
 - iii. 2 - Good color all over.
 - e. Respiration (Breathing):
 - i. 0 - Not breathing.
 - ii. 1 - Weak cry; may sound like whimpering, slow or irregular breathing.
 - iii. 2 - Good, strong cry; normal rate and effort of breathing.
- b. Initiate cardiac monitoring, pulse oximetry, and serial vital signs.
 - 1. Oxygen saturations rise gradually over the first ten minutes of birth, relative normal O2 sats are listed below:
 - a. 1 min 60%-65%
 - b. 2 min 65%-70%
 - c. 3 min 70%-75%
 - d. 4 min 75%-80%
 - e. 5 min 80%-85%
 - f. 10 min 85%-95%
 - ii. Reassess for neonatal distress; gasping respirations, HR < 100, central cyanosis.
 - 1. If none, encourage bonding with mother/nursing prior to, but not during, transport.
 - 2. If any distress noted proceed to resuscitation per NRP Guidelines
- c. Resuscitation
 - i. Provide warmth to the infant.
 - 1. Utilize external porta-warm mattress while avoiding direct contact with the infant's skin.
 - ii. Dry the baby and stimulate breathing.
 - iii. Simultaneously evaluate respirations, heart rate, and color. Perform initial APGAR score.
 - 1. If apneic or HR < 100 bpm, begin positive-pressure ventilations.
 - 2. If breathing and HR > 100 bpm but with central cyanosis, administer supplemental oxygen.
 - 3. If breathing, HR > 100 bpm and pink, observation and continuous evaluation is indicated.
 - iv. After 30 seconds, reassess for apnea or HR < 100/bpm:
 - 1. If positive-pressure ventilations are effective and HR remains > 100/bpm, proceed stabilization
 - 2. If labored breathing/cyanosis persists but HR > 60 bpm consider CPAP

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3. If despite adequate ventilation with supplemental oxygen the HR < 60 bpm, continue positive-pressure ventilations & begin chest compressions at rate >100 bpm.
- v. Most neonates should **not** require endotracheal intubation. For prolonged resuscitation consult with receiving physician or medical control to consider tracheal intubation per *GUID 3203-PR020 Endotracheal Intubation* with the following caveats:
 1. The provider with the most experience intubating neonates performs the task
 2. Use of RSI medications are generally not indicated in the neonatal population.
 3. Choose cuffed tracheal tubes of appropriate size for patient:
 - a. 2.5 TT for < 1,000 g and below 28 week gestation.
 - b. 3.0 TT for 1,000 - 2,000 g (1 - 2 kg) and 28 - 34 weeks.
 - c. 3.5 TT for 2,000 - 3,000 g (2 - 3 kg) and 34 - 48 weeks.
 - d. 3.5 - 4.0 TT for > 3,000 g (3 kg).
 4. For infants over 34 weeks old an appropriately sized LMA or iGel may be used
 5. Examine the patient for indications of congenital airway or maxillofacial anomalies that may make intubation difficult (e.g., micro or retrognathia, macroglossia, etc.). If airway anatomy is suspected or confirmed to be abnormal discuss with Medical Control.
 6. Place pulse oximetry preferably on right hand and cardiac monitor.
 7. After intubation, maintain the tube's position by holding the ring finger firmly over the tube on the roof of the infant's mouth.
 8. Confirm correct depth of tube.
 - a. 6 cm mark at teeth: < 1,000 g
 - b. 7 cm mark at teeth: 1,000 - 2,000 g
 - c. 8 cm mark at teeth: 2,000 - 3,000 g
 - d. 9 cm mark at teeth: 3,000 - 4,000 g
 - e. 10 cm mark at teeth: 4,000 g
 9. Secure the endotracheal tube with commercial device or tape.
 10. Reconfirm appropriate position of tracheal tube.
 11. After TT is secure, or if prolonged BLS manual ventilation is required, place an OG tube to decompress the stomach.
- vi. Establish IV/IO access.
- vii. Consider **Naloxone 0.1 mg/kg IV/IO** per NRP guidelines ONLY if patient cannot be adequately ventilated.
- viii. After 30 seconds, reassess HR:
 1. If HR remains < 60 bpm despite adequate ventilation and chest compressions:
 - a. Administer **Epinephrine 0.01 mg/kg IV/ IO** (1 mg/10 mL (0.1 mg/mL))
 - b. If **no IV/IO then administer Epinephrine 0.1 mg/kg** (1 mg/10 mL (0.1 mg/mL)) via the tracheal tube, maximum volume 3 mL.
 - c. Repeat Epinephrine dose every 3 - 5 minutes as needed.
 - d. Consider NS bolus of 10 mL/kg IV/IO.

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ix. If at any time resuscitation is effective and HR remains > 100 bpm, proceed to stabilization

V. **DOCUMENTATION:** EMS Charts

VI. **REFERENCES:**

a. **APGAR Scoring**

	Sign	0 Points	1 Point	2 points
A	Activity (Muscle Tone)	Absent	Arms and legs flexed	Active movement
P	Pulse	Absent	Below 100 bpm	Above 100 bpm
G	Grimace (Reflex Irritability)	No response	Grimace	Sneeze, cough, pulls away
A	Appearance (Skin Color)	Blue-gray, pale all over	Normal, except for extremities	Normal over entire body
R	Respiration	Absent	Slow, irregular	Good, crying